

Varuna FloodTwin: An Open Urban Flood Digital Twin

from Free Satellite Data, and an Honest Measurement of What Its Accuracy Score Is Worth

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Abstract—Indian cities flood every monsoon, and most have no calibrated flood model to plan with. We present Varuna FloodTwin, an open pipeline that builds a flood model of any place on Earth from free satellite data, runs fast enough to answer questions live in a web page, and is deployed for 16 city areas in India entirely on free computing tiers. Our main claim is not that the model is accurate. It is that we measured what such accuracy scores are worth. On 150 Sentinel-1 radar scenes across 15 areas, a baseline with no physics and no rainfall, which simply predicts the cells wet in most other storms, scores a Critical Success Index of 0.29 to 0.67. That beats every physics-model score we can find published for this task, including our own, and in all 15 areas it beats the agreement between two radar passes of the same city. A learned model on the same terrain rasters scores 0.288 and ties that baseline to three decimal places, so its value is portability rather than accuracy. Held out one whole region at a time it scores 0.147, which a threshold correction that never sees a test label raises to 0.175, and adding a single coastal city to training lifts held-out coastal Mumbai by 84 %, turning a suspected hard limit into a data-coverage instruction. Model size, grid resolution and free reanalysis rainfall all change nothing. Planning nonetheless survives this location uncertainty, because every intervention is valued by re-simulation. A dense storm drain network routed on the real street graph cuts simulated street flooding by 80.9 % in Patna at a lower cost per cubic metre than sparse canals, and a metered recharge planner banks 6.3 to 16.5 % of storm runoff in six depleted Karnataka towns against 0.7 to 3.3 % on the flood plain.

Index Terms—Urban flooding, digital twin, Sentinel-1, flood validation, managed aquifer recharge, open data.

I. INTRODUCTION

PLUVIAL flooding, meaning flooding caused by rain falling on the city itself rather than by a river bursting its banks, is one of the most frequent climate impacts in South Asian cities. The planning reality in most Indian municipalities is a drainage map older than the city that grew around it, no calibrated hydraulic model, and no budget to buy one. A tool that could change decisions there must be buildable from free global data, because nobody will survey the city for us, cheap enough to run interactively, because planning is a conversation and not a batch job, and honest about what it cannot claim, because a confident wrong answer about which street floods is worse than no answer. Fig. 1 is what a planner sees.

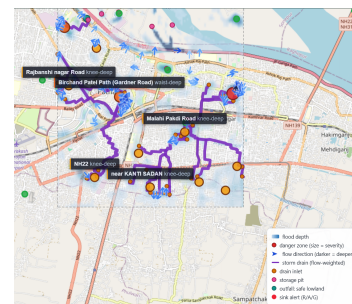


Fig. 1. The live system for Patna at a 100 mm design storm. Blue is predicted depth, arrows show flow, purple lines are the drain network the planner designed on the real street graph, and labels name real streets with a depth in words, because “waist-deep” is a decision and “0.94 m” is only a measurement.

Related work. Our numerical core is the local inertial shallow water scheme of Bates *et al.* [1], re-implemented in PyTorch so that every simulation is differentiable end to end, and near-concurrent differentiable solvers [2] ship no emulator, hosted twin or planning stack. Flood Hub’s global river-flood forecasts [3] are the closest deployed system and the river counterpart of what we do for rain-driven city flooding, while graph networks for flooding put hydraulics into message passing on a mesh [4], whereas our graph is the road network itself, so what gets predicted is the flooding of a named street. Wing *et al.* [5] set the standard for validation against observations, Marsh *et al.* [6] give the elevation error budget behind our uncertainty analysis, and managed aquifer recharge is studied as a joint answer to floods and groundwater depletion [7]. What we could not find anywhere is a paper reporting its accuracy score beside a trivial baseline computed on the same scenes, and Section III argues that this omission makes most reported flood scores impossible to interpret.

II. THE SYSTEM

For any area, Google Earth Engine exports four free global datasets. FABDEM [8] gives ground elevation at 30 m with forests and buildings stripped out, so the surface is what water actually runs over. ESA WorldCover [9] gives land cover at

TABLE I

WHAT A CSI IS WORTH ON SENTINEL-1 FLOOD EXTENT, WITH NO MODEL AT ALL. 150 RADAR SCENES ACROSS 15 AREAS, OF WHICH 149 ARE SCORED. MULTI-AREA ROWS ARE MEANS, OVER PER-AREA CLIMATOLOGY RANGES OF 0.39 TO 0.60 FOR MUMBAI AND 0.52 TO 0.67 FOR KARNATAKA.

region	areas	storms	all-wet	clim.	SAR/SAR
Patna (inland)	1	7	0.054	0.286	0.214
Bengaluru (inland)	1	10	0.022	0.577	0.502
Mumbai (coastal)	6	62	0.019	0.520	0.426
Chennai (coastal)	1	10	0.021	0.500	0.383
Karnataka (inland)	6	60	0.021	0.615	0.532
all	15	149	0.022	0.545	0.456

10 m, JRC surface water [10] marks permanently wet cells so a lake is not mistaken for a flood, and Sentinel-1 radar [11] gives observed water on real storm dates, because radar sees through the cloud that a monsoon guarantees.

The domain is a square grid at 60 m, using 128^2 cells for Patna and Bengaluru and 256^2 for each Mumbai tile. Mumbai tiles share exact edges so summing them never double-counts ground, and they are independent models with closed boundaries, so tides are not simulated and the dashboard says so. The solver is the Bates scheme in PyTorch, so we can ask what change in a physical parameter would have changed the flood and answer with a gradient rather than by trying values one at a time. Land cover maps to Manning roughness and to infiltration, which is metered against a finite soil storage capacity and capped at the saturated conductivity of the soil. A U-Net [12] imitates the solver so the dashboard answers what-if questions instantly, with wet cells carrying twenty times the loss weight because flooding is rare in space, reaching 5 to 21 cm validation error. A FastAPI backend serves committed per-city bundles including cached OpenStreetMap road graphs [13], so the deployed service needs no external API at run time, and a new city window takes 15 to 25 minutes to build on a free T4 GPU.

III. WHAT A FLOOD ACCURACY SCORE IS WORTH

The most useful thing we can report is not a model score. It is the range a flood score can occupy on this task *before any model exists*.

The Critical Success Index (CSI) is the standard overlap measure in flood mapping, being the number of cells the model and the radar agree are wet divided by the number either one calls wet. Three baselines cost nothing and bracket it. *All-wet* predicts every cell is flooded, so its CSI is exactly the observed wet fraction, earned with zero knowledge. *Random* scatters the correct number of wet cells uniformly. *Climatology* predicts the cells wet in most *other* storms of that same area, using no rainfall, no physics and no elevation, with one threshold fixed at 0.5 for every area rather than tuned per area. We also score every radar date of an area against every other date of it, which is not a baseline but an empirical ceiling, because it measures how much two radar passes of one city agree.

Table I is the answer. A climatology with one fixed threshold and no other parameter scores 0.29 to 0.67 per area, with a

TABLE II

ALL METHODS ON THE IDENTICAL 18 STORMS, EVERY THRESHOLD CHOSEN OFF THE REPORTED SCENE. THE LEARNED MODEL IS A 7.9 M-PARAMETER U-NET OVER 23 TERRAIN CHANNELS WITH RAINFALL REMOVED, AVERAGED OVER 3 SEEDS.

method	mean CSI	$\times$ all-wet
random	0.014	0.5
all-wet	0.029	1.0
best static index (HAND, TWI)	0.039	1.3
dynamic twin (antecedent rain)	0.054	1.9
learned (terrain only)	0.288 ± 0.007	9.9
climatology (no model, no rain)	0.288	9.9

mean of 0.545. That is higher than any CSI we are aware of being reported for a physics-based urban flood model on Sentinel-1, including our own. It is also higher, in all fifteen areas individually, than the agreement between two Sentinel-1 passes of that same city. Averaging many passes is more self-consistent than any two of them, which is a property of the target and not of a model, and it caps what any model can honestly be asked to do here.

The level is so high because the Sentinel-1 target is dominated by water that has nothing to do with the storm and that the permanent-water mask does not remove, being tidal flat, mangrove, seasonal pond and paddy field. CSI on this task has therefore mostly been measuring *can you find standing water*, which our twin never attempted, because it predicts only the water the storm adds. The signature confirms this. On Mumbai-Northeast’s one genuine flood, with 2937 wet cells against a baseline near 450, the climatology scores its *worst* value of 0.075. It wins the quiet dates and loses the flood. CSI in this literature should therefore always be reported beside all-wet and a leave-one-out climatology on the same scenes. Without them the number is uninterpretable, and in retrospect that applied to the 0.041 we published ourselves.

IV. THE TWIN RE-SCORED, AND WHERE THE INFORMATION IS

Sentinel-1 masks for 18 monsoon storms across Patna and Mumbai-Northeast are scored on a common 60 m grid with permanent water masked *on both sides*. Every method sees the identical grid, mask and target, and every threshold is chosen on the *other* storms of the fold and never on the scene reported.

Table II carries two results. First, the same elevation, land cover and soil rasters, read by a U-Net instead of routed by a solver, reach 0.288 ± 0.007 , which is 5.4 times the twin on identical scenes and refutes our own earlier claim that terrain fundamentally lacks the information. What fails in this regime is *routing* water over a noisy elevation model, not *predicting* where water goes. The feature that makes the difference is available to both methods, being elevation minus its own blurred copy at 2, 8 and 32 cells. Elevation error is spatially correlated, so neighbouring cells are wrong together and by similar amounts, and this relative micro-relief survives a bias that destroys absolute elevation.

Second, the learned model scores 0.2884 and the climatology scores 0.2881, a gap of 0.0003 against a seed standard

deviation of 0.0066. On a city with a radar archive, a neural network over 23 terrain channels is worth nothing over the statement “these cells are usually wet”. This is the hinge of the paper. The learned model’s contribution is not accuracy but portability.

Why the physics router fails. Under a ten-member ensemble of spatially correlated elevation perturbations at FAB-DEM’s nominal 1 m error class [6], total flooded area for Patna at 100 mm is robust at $10.79 \pm 0.35 \text{ km}^2$, but only 5 % of it floods in at least 90 % of members, against 52 % in ten times hillier Bengaluru. Calibrating 16 per-land-class roughness and infiltration multipliers by gradient descent moves held-out CSI only from 0.0483 to 0.0488, because those parameters change *how deep* the water is, not *where* it is. Where metre-scale elevation noise exceeds the relief that decides which street ponds, a solver that routes water downhill cannot place it.

V. PORTABILITY, COVERAGE, AND A FREE CORRECTION

If a learned model ties a parameter-free climatology on a city with a radar archive, the only reason to prefer it is that it runs where the climatology cannot be computed at all. A city with no Sentinel-1 history has no climatology, so the question that matters is unseen ground.

Holding out a tile is not holding out a city. Leaving out one *tile* gives 0.270 ± 0.015 , which is 12.5 times all-wet and misleading. With six Mumbai tiles and seven Karnataka areas in the dataset, holding out Mumbai-West leaves five adjacent Mumbai tiles in training, and holding out Chitradurga leaves six other Karnataka towns sharing its climate and tank morphology. That is generalisation to unseen *ground*, not to an unseen city. Holding out an entire region instead gives 0.147 ± 0.008 , half the per-tile figure and 6.9 times all-wet, and it is the number we report as deployable. Because we also rewrote the dataset builder between our earlier three-area figure of 0.094 ± 0.021 and this one, we ran those three areas on the rebuilt stack as a control. It gives 0.096 ± 0.009 , so the rebuild moved nothing and the gain is areas. Transfer is not uniform, which is the operationally important part. Held out entirely, the seven Karnataka areas score 0.213 at 10.3 times all-wet and Patna scores 0.166, while the six Mumbai tiles score 0.033, close to predicting nothing. Mumbai’s radar water is tidal flat, creek and mangrove, and neither the Gangetic flood plain nor the Karnataka plateau contains one example of that surface. Either coastal ground needs coastal training data, or tidal water is not predictable from terrain at all.

The coastal limit is coverage, not unlearnability. To separate those readings we built a fifteenth area. Chennai is 1300 km from Mumbai, shares no catchment with it, and is kept as its own region on purpose, so that Chennai’s fold trains on a set containing Mumbai while Mumbai’s fold trains on a set containing Chennai, and one experiment tests both directions at once. Building it exposed a defect worth reporting. Our scene selector had the monsoon window hard-coded to 1 June through 15 October, but India has two monsoons and Chennai floods in the northeast monsoon from October to December, so the scenes we could even ask for

were dry-season ones. After the fix the wettest available date went from 40.3 to 180.5 mm of antecedent rain and one date turned out to have five to ten times the wet fraction of any other Chennai scene, an actual flood the first selection could not see. A season window is a scientific parameter and not a configuration detail.

The result is that held-out Mumbai goes from 0.033 to 0.061, and Chennai scored with Mumbai in training reaches 0.164, which is 7.8 times all-wet, close to inland Karnataka’s 10.6 and nothing like the 1.8 Mumbai managed as the only coast in the dataset. Tidal water is learnable from terrain. The model simply has to have seen a coast. The experiment carries its own control, because every fold gained the same ten storms and only the coastal region moved, at +84 % on the coast against +3.6 % for Karnataka and −7.4 % for Patna, lifting the overall figure from 0.131 to 0.147. For scale, going from 3 to 14 areas was 4.6 times the storms for +39 % overall, whereas this was +13 % more storms for a doubling of one region. The rule that follows is concrete. Deploy on an unseen inland city freely, and on an unseen coastal city only with a coast in training.

A threshold correction that costs nothing. In every experiment here the model’s ranking quality runs above its achieved CSI, which is calibration loss. The fix is free. For each scene, pick the threshold that makes the *predicted* wet fraction match the mean observed wet fraction of the *training* scenes, a prior that never looks at a test label. On the leave-one-region-out split, which is the deployment question, this lifts CSI from 0.1474 to 0.1746 at 15 areas and from 0.1305 to 0.1566 at 14, which is 3.4 and 5.7 seed standard deviations and captures 73 and 84 % of the gap to an oracle threshold chosen knowing the test scene. On the per-tile split the same correction *hurts*, at 0.2698 to 0.2542, and that sign flip is the mechanism rather than noise, because holding out one tile leaves five sibling tiles in training, so the training prior already describes it, whereas an entire held-out region is genuinely miscalibrated. It therefore helps most exactly where the model is weakest, gaining +73 % for coastal Mumbai against +5.8 % for Karnataka, because at the global threshold the model finds only half to four-fifths of the water actually there, and under-warning is the wrong direction for a flood product. Held-out coastal Mumbai has now moved from 0.033 with no coast in training, to 0.061 with one coast, to 0.106 with the free correction. That is 3.2 times the starting figure, and neither step needed a larger model or a finer grid. One was data coverage and the other was arithmetic.

VI. THREE LEVERS THAT ARE NOT LEVERS

When a model underperforms, the reflex is to make it bigger, give it a finer grid, or feed it more inputs. We tried all three and report the negative result, because it saves other people the same weeks. **Capacity does nothing.** On held-out floods with terrain only, CSI is 0.221 at 2.0 M parameters, 0.232 at 7.9 M, 0.221 at 31.4 M and 0.219 at 70.6 M, flat across a 35-fold range and inside seed noise. **Resolution does nothing.** Re-running at 30 m quadruples the cell count and changes nothing, at 10.4

times all-wet against 9.7 at 60 m, the multiple being the column to read because raw CSI is not comparable across grids.

Rainfall is not usable and actively hurts. Removing the rainfall channels *improves* ranking quality by 0.124, about seven seed standard deviations with the three-seed ranges disjoint. Shuffling which rainfall vector goes with which storm, which preserves every distribution and destroys only the pairing, lands monotonically between real rain and no rain, at 0.184 then 0.272 then 0.313, so real rainfall is strictly the worst of the three inputs. Two alternative explanations were tested and refuted, being tidal contradiction between Mumbai tiles and failure to extrapolate to larger storms. On an unseen region rainfall neither helps nor hurts, at 0.1460 against 0.1474, so the claim is not that rainfall is universally harmful. It is that rainfall lets the model take a shortcut on a city it has already seen, and that shortcut does not survive the move to new ground. There is nothing to extract because nine probe points spanning a 15 km tile collapse to a single archive cell. Spatial rainfall is not merely unused. It is unavailable.

VII. PLANNING THAT SURVIVES LOCATION UNCERTAINTY

Section IV says we cannot reliably name the street that ponds first, but planning is still possible, because the quantities that drive a plan are not the quantity that fails. Every intervention is evaluated the same way. Modify the terrain, re-simulate the design storm at 100 mm over 24 h, and measure flooding on built-up cells, *excluding* the intervention's own cells from both masks, because moving water onto a drain does not count as removing it.

A storm drain spiderweb on the real street graph. Sparse worst-basin-to-nearest-outfall canals are how our earlier planner worked, and anecdotally how much real retrofitting works. They cut 20.8 % of street flooding in Patna and 25.7 % in Bengaluru. Scaling the idea up, we route a dense network on the actual street graph, whose nodes carry elevations, with 42.8 k nodes for Patna, and whose edge cost is street length plus an uphill penalty of 2400 m per metre of climb. One reverse multi-source Dijkstra search from all outfalls labels every street with its gravity-cheapest exit, about 40 inlets are placed at the deepest cell of each flood basin, and their exit chains trace a forest with shared trunks that accumulate flow. The outfalls encode a safety principle, which is *never the river*, because during a flood the river runs high and a drain into it runs backwards, so we use the lowest *safe* ground instead. The chosen streets are carved into the elevation as 2 m channels with descending beds, and the storm is re-simulated.

Across the eight built areas the spiderweb cuts street flooding by 80.9 % in Patna, 89.4 % in Patna-East, 80.3 % in Patna-West, 72.6 % in Bengaluru, and 63.0 down to 32.5 % on the four largest Mumbai tiles, against 21 to 26 % for the sparse plans, with monotone dose-response over 25 to 200 mm. For Patna that is 0.87 million m³ of flood water removed against 0.28 million for the sparse plan, and density improves the economics as well as the effectiveness, at INR 850 per cubic metre removed against INR 1 309, because shared trunks spread their length across many basins. For comparison, excavation alone

at eight sites cuts 4.1 %, detention pits alone cut 7.8 %, and 727 distributed storage basins cut 30 % at INR 9 200 per cubic metre. The lower Mumbai cuts are informative rather than disappointing, because those tiles are four times larger and partly harbour, so a fixed budget of 40 inlets covers a smaller share of the flooding. A separate storage planner ranks detention tanks by pooled depth at dynamic local minima, excluding building footprints because you cannot dig a tank under an occupied house. It has never seen a flood report or a news archive, yet its top-ranked sites land on each city's chronically flooded places, including Koramangala in Bengaluru and the Parel and Byculla belt in Mumbai where the municipal corporation actually built its underground tanks.

From flood to aquifer, metered rather than assumed. The water that floods an Indian city in July is water the same region is short of in April. Most recharge plans assume an infiltration rate and multiply. We can do better, because the model already meters infiltration against a finite soil capacity, so we build recharge structures into it, re-simulate the storm, and read off how much water actually went underground. Choosing where to build a twin is itself a decision, so we screen an entire state first, combining Central Ground Water Board stage-of-extraction, a GRACE-assimilated storage trend, HydroSHEDS flow accumulation and SoilGrids conductivity over Karnataka's 27 district units, which ranks Bangalore Rural, Chitradurga, Chamarajanagara, Kolar and Bangalore Urban first at 111 to 187 % extraction, all officially over-exploited or critical. The measurement across fourteen built areas then splits cleanly. The six Karnataka towns send 6.3 to 16.5 % of a 100 mm storm's runoff into the aquifer, led by Chikballapur at 607 172 m³ and Kolar at 667 548 m³, while the flood plain and the coast manage 0.7 to 3.3 % and Mumbai-South is indistinguishable from zero. That gap is what the screen predicted and it has a physical reason. On a flood plain the water table is already close to the surface and the soil column is nearly full, so there is nowhere to put the water, whereas on the Karnataka plateau the aquifer is mined and hard rock sits under thin soil, so the storage is real and empty. The same structures cut flooding by 46.9 to 88.2 % in those towns, so this is no trade between flood control and water supply. Where recharge works, it is the same intervention.

An external anchor from inferred drainage. The largest known gap in the twin is that no storm sewer is modelled, so it over-floods precisely where working drains exist. Mumbai will not hand out a drainage map, so we fit a per-cell outflow field through the checkpointed simulator on 8 storms and hold out 3, with a minimum-outflow prior charging for every cubic metre drained, so the fit picks the *smallest* drainage explaining the observed extent and every volume is a lower bound. Held-out extent improves by 10 % while predicted wet cells *fall* by 10 %, the opposite of the usual trade where CSI is bought by predicting more water. Integrating the drained volume gives Mumbai's whole-city surface outflow as at least 3.48 million m³ at 25 mm rising to only 5.41 million at 200 mm, so outflow saturates while flooded area keeps climbing, a falsifiable ceiling near 5.4 million cubic metres per storm

however hard it rains. The municipal corporation reported pumping about 4.7 million m³ per day through six pump houses over 16 to 19 August 2025, and a total-outflow lower bound sitting just above a pump-only throughput is the right side to sit on, though this is a consistency check and not a validation.

Learned street-level risk and routing. A planner asks where to build, but a person standing in the rain asks which way to walk. A message-passing network over the street graph, conditioned on rainfall, distils the twin into per-street risk at 0.890 to 0.942 AUC on held-out storm sizes in about 4 ms against 260 ms for the raster pipeline, and held out of Bengaluru entirely it still ranks that city’s flooded streets at 0.721 AUC, the same coverage lesson the coastal experiment taught. Routing on the inflated cost, judged on oracle depths over 150 random trips, recovers 64.6 % of the oracle’s floodwater avoidance in Patna for 57 % of its detour.

VIII. LIMITATIONS AND CONCLUSION

The hydraulics run at 60 m, coarser than the streets they describe. Against 83 quarantined crowdsourced depth reports from the public mumbai.flood.in service, the twin underpredicts depth by a stable factor of 4.45, implying a real ponding footprint about 28.5 m across, so a finer grid is the clearest next step and it has a quantitative target, while correlation with observed depth there is near zero, the same co-location failure the radar work found, now confirmed by an independent instrument. The evidence base is uneven, because Table II rests on 18 storms across two areas while everything after it rests on 15 areas and 150 scenes. The Mumbai tiles are closed-boundary models, so seam flow and tidal coupling are absent, and the coastal result rests on two cities. Recharge volumes come from metered infiltration and stand on their own, but within-town site rankings use placeholder groundwater data, and recharge has a water quality dimension a flood model cannot see. Costs are indicative and rank strategies rather than price projects. This planner prioritises. It does not certify.

What we would keep from all this is one habit. A flood overlap score on this task is uninterpretable on its own, because a baseline with no physics, no rainfall and one fixed parameter scores 0.29 to 0.67 across 15 areas, beats every physics-model score we can find published, and beats the agreement between two radar passes of the same city in all fifteen. Any CSI in this literature, ours included, should be read next to those numbers or not read at all. Once that scale is fixed the rest changes shape. Our physics twin scores 0.054 and a learned model on the identical rasters scores 0.288, refuting our own earlier claim that terrain fundamentally lacks the information, but that model ties the parameter-free baseline exactly, so its value is not accuracy. Its value is that it runs on cities with no radar history at all, worth 0.147 held out one whole region at a time and 0.175 with a free threshold correction, and the one place it nearly fails is a coast it has not seen, which is coverage rather than a hard limit. Meanwhile planning works despite that uncertainty, because every intervention is measured by re-simulation instead of assumed. We have reported the failures

beside the successes throughout, because we would rather the record be useful than flattering.

REPRODUCIBILITY AND ACKNOWLEDGMENT

Code, all per-area artifact bundles including the Sentinel-1 water masks, road graphs and ranked site lists, with offline tests and runbooks, are public at <https://github.com/Maverick-Ansh/project-varuna>, and the live dashboard is at <https://project-varuna-iota.vercel.app>. Every headline number here is regenerable from a committed artifact, and the headline is unchanged on a fully rebuilt dataset, at 0.288 ± 0.008 against 0.288 ± 0.007 . This work uses FABDEM, ESA WorldCover, JRC Global Surface Water, Copernicus Sentinel-1, OpenStreetMap, Open-Meteo, SoilGrids, HydroSHEDS and Central Ground Water Board data, all free to use, plus crowdsourced observations from mumbai.flood.in, and runs on free tiers from Google Colab, Kaggle, Hugging Face and Vercel.

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